

Igor Lima Rocha Azevedo

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EDUCATION

Imperial College London Sep 2025 - Sep 2026

Master of Research (MRes) in Artificial Intelligence and Machine Learning

- Conducting research on generative and contrastive deep learning for predicting rare events. Supervised by Prof. Pedro Mediano

University of Brasilia Aug 2016 - Jun 2022

Electrical Engineering

- Conducted research on FPGAs, as well as Machine Learning for audio codec optimization and financial markets. Supervised by Prof. Edson Mintsu Hung.

RELEVANT EXPERIENCES

Research Scholar at The University of Tokyo Tokyo, Japan | Apr 2023 - Apr 2025

- Contributed to deep learning research projects supervised by Prof. Toyotaro Suzumura and Dr. Yuichiro Yasui.
- Collaborated with Nikkei Inc. (a Financial Times group company) to design novel recommender system models.
- Developed deep learning models for high-frequency stock price forecasting.
- Researched foundational LLMs, exploring retrieval-augmented generation (RAG) and long-context handling.

Developer & Development Team Lead at VOGA Brasilia, Brazil | Jul 2021 - Apr 2023

- Started as a Developer, later promoted to Development Team Lead.
- Led system integration for VOGA post-acquisition by BTG Pactual, South America's largest investment bank.
- Architected and developed a centralized platform to monitor and track over USD 300 million in managed assets.
- Transitioned from full-stack development (Flask/NextJS) to a DevOps role, managing AWS infrastructure (EC2, VPC, RDS, ELB) with Terraform, automating CI/CD with Jenkins, and securing the application with Cloudflare.

Research Intern at Cellcrypt: Quantum-Safe Encrypted Calls London, UK (Remote) | Sep 2020 - Jun 2021

- Optimized machine learning models to improve VoIP performance by enhancing call quality and reducing latency.
- Built an ML pipeline for automated PJSIP (multimedia communication library written in C with high level API in C++) parameter tuning, which boosted call quality metrics by 7%.

SELECTED PUBLICATIONS

A Look Into News Avoidance Through AWRS: An Avoidance-Aware Recommender System Jul 2024
SIAM International Conference on Data Mining (SDM'25)

- A novel architecture for learning from and modeling user avoidance patterns in recommendations.
- Achieves better performance on key metrics across multilingual datasets in Japanese, English, and Norwegian.

NewsReX: A More Efficient Approach to News Recommendation with Keras 3 and JAX Aug 2025
Under Review

- An open-source JAX-based library for reproducible state of the art news recommendation models.
- Achieved up to 41% faster model training for consumer-grade GPUs.

A SHA-3 Co-Processor for IoT Applications Nov 2020
IEEE Workshop on Communication Networks and Power Systems (WCNPS'20)

- Implements a SHA-3 Co-Processor in FPGA using C++ suitable for IoT applications.
- Proposes solution is about 65% faster than the ARM Cortex-A9 when oputing SHA-3.

AWARDS

Society for Industrial and Applied Mathematics (SIAM) Travel Award May 2025

- Recognizing promising early-career researchers in the field of data mining.

Japanese Government Research Scholarship April 2023 - April 2025

- A prestigious and highly competitive scholarship to support students conducting research at higher education institutions in Japan.

Brazilian Government Scientific Initiation Scholarship August 2019 - July 2020

- A federal government-funded program to support undergraduate students engaging in research and innovation.

LANGUAGES & SKILLS

Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate), Japanese (Beginner)

Technical Skills: Python, PyTorch, TensorFlow, Keras, JAX, SciKit-Learn, Lightning AI, CrewAI, DSPy, Flask, Java, C, C++, SQL, AWS (EC2, ECS, RDS, VPC, S3), Terraform, Linux, and MATLAB.

Soft Skills: Cross-Functional Collaboration, Adaptability, Analytical Thinking, and Attention to Detail.